



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SARADHA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

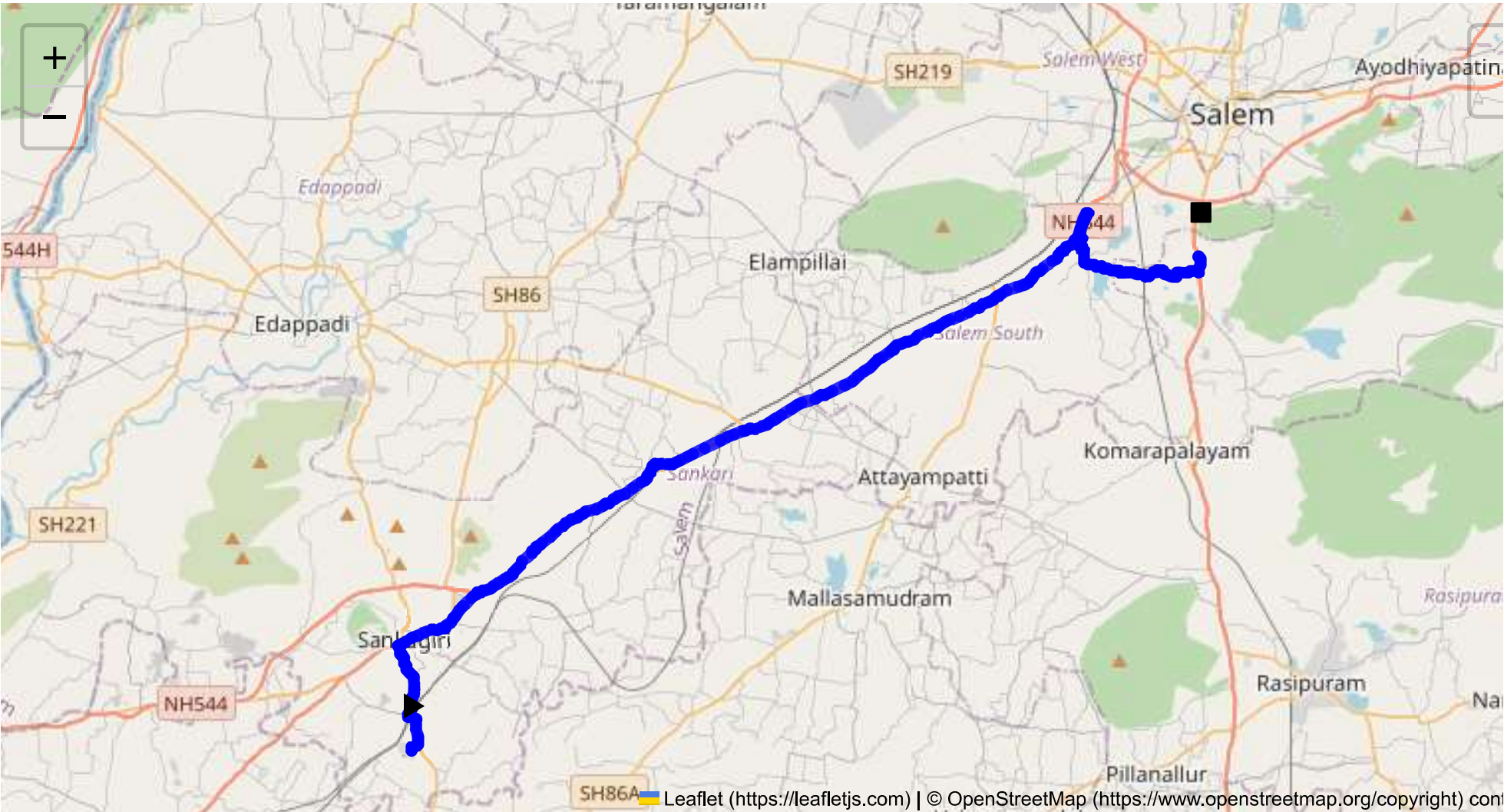
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 43.98 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.606636, 78.146238)



Welcome to the Journey Risk Management Study

Certainly! To provide a detailed analysis for the specified route from CVQF+23W, Sangagiri to J44W+GJC, Dasanaickenpatti, here is a comprehensive overview focused on safety and potential hazards for truck drivers, particularly those carrying hazardous materials:

1. Overview of the Route Map:

- The route spans approximately 44 kilometers through the Tamil Nadu region of India, linking Sangagiri to Salem. The journey typically involves traveling along NH 544 (National Highway 544), a major roadway that is commonly used for regional and intercity travel.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu experiences a tropical climate with distinct wet and dry seasons. The southwest monsoon brings heavy rains from June to September, potentially causing road flooding and reduced visibility. The northeastern monsoon in October-December can also result in similar conditions. Drivers should be cautious of rain-induced slippery roads during these periods.

3. Traffic Patterns:

- Peak traffic hours typically occur during the morning (7:00-10:00 AM) and evening (5:00-8:00 PM). Congestion-prone areas include the approaches to larger towns and cities such as Salem, as well as junctions and toll plazas along NH 544.

4. Assessment of Road Quality and Infrastructure:

- NH 544 is generally well-maintained, but varying road quality can be expected near urban centers and some rural stretches. Recent or ongoing infrastructure projects might cause temporary detours or lane closures.

5. Suggestions for Alternative Routes for Emergencies:

- In case of major disruptions on NH 544, state highways like SH 86 can serve as detours, though they might add to travel time and should be assessed for suitability of truck traffic.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Compliance with Tamil Nadu's transport regulations is crucial. This includes proper labeling, restricted timings for transport, and specific permits for hazardous materials. Consultation with local transport authorities is advised for up-to-date regulations.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- Historical data highlight accidents primarily due to overloading, driver fatigue, and mechanical failures. Known incident hotspots include areas with sharp curves or inadequate signage.

8. Environmental Considerations and Sensitive Areas:

- Be mindful of passing through buffer zones of any wildlife sanctuaries or protected areas. Restrictions may apply to reduce environmental impact.

9. Analysis of Communication Coverage:

- Mobile network coverage is mostly sufficient along NH 544 but can become spotty in rural stretches between town clusters. Dead zones may occur, so preemptive communication planning is advised.

10. Estimated Emergency Response Times for Different Route Segments:

- Near urban centers like Salem, emergency response times could be as short as 15-30 minutes. In rural areas, response times may extend to 45 minutes or more due to distance and road access.

11. Overall Summary of Risk Assessment:

- The route poses moderate risk due to weather conditions during monsoons, peak traffic congestion, and occasional infrastructure quality discrepancies. With proper precautions, adherence to regulations, and awareness of conditions, safety can be effectively managed for hazardous material transport on this route.

This comprehensive analysis should serve to guide and train truck drivers, equipping them with the knowledge needed to navigate the route safely and efficiently while minimizing risks associated with heavy vehicle transport.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45004, 77.87206	15 KM/Hr
5	Turn	Medium	11.45028, 77.87202	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	Medium	11.47410, 77.86845	30 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
11	Turn	High	11.62130, 78.10776	15 KM/Hr
12	Blind Spot	Blind Spot	11.61217, 78.10439	10 KM/Hr
13	Blind Spot	Blind Spot	11.61275, 78.10580	10 KM/Hr
14	Turn	Medium	11.61055, 78.10525	30 KM/Hr
15	Turn	High	11.60904, 78.10584	15 KM/Hr
16	Turn	Medium	11.60869, 78.10634	30 KM/Hr
17	Turn	Medium	11.60618, 78.10646	30 KM/Hr
18	Turn	High	11.60456, 78.10638	15 KM/Hr
19	Turn	Medium	11.60448, 78.10648	30 KM/Hr
20	Turn	Medium	11.60311, 78.11108	30 KM/Hr
21	Turn	High	11.60169, 78.11880	15 KM/Hr
22	Turn	Medium	11.60139, 78.11899	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
23	Turn	Medium	11.60094, 78.12130	30 KM/Hr
24	Turn	Medium	11.60106, 78.12149	30 KM/Hr
25	Turn	Medium	11.59979, 78.12906	30 KM/Hr
26	Turn	Medium	11.60190, 78.13419	30 KM/Hr
27	Turn	High	11.60182, 78.13428	15 KM/Hr
28	Turn	Medium	11.60094, 78.13434	30 KM/Hr
29	Turn	Medium	11.59968, 78.13855	30 KM/Hr
30	Turn	Medium	11.60127, 78.13924	30 KM/Hr
31	Turn	High	11.60135, 78.13930	15 KM/Hr
32	Turn	Medium	11.60085, 78.14239	30 KM/Hr
33	Turn	Medium	11.60195, 78.14409	30 KM/Hr
34	Blind Spot	Blind Spot	11.60119, 78.14658	10 KM/Hr
35	Blind Spot	Blind Spot	11.60678, 78.14652	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
19	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
20	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
21	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
22	school	Sugam Matric Higher Secondary School	11.6069088, 78.1462056	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45004, 77.87206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47410, 77.86845



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49149, 77.89601



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49224, 77.89601



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62145, 78.10740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61217, 78.10439



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.61275, 78.10580



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61055, 78.10525



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60904, 78.10584



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60869, 78.10634



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60618, 78.10646



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60456, 78.10638



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60448, 78.10648



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60311, 78.11108



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60169, 78.11880



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60139, 78.11899



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60094, 78.12130



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60106, 78.12149



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59979, 78.12906



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60190, 78.13419



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60182, 78.13428



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60094, 78.13434



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59968, 78.13855



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60127, 78.13924



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60135, 78.13930



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60085, 78.14239



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60195, 78.14409



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.60119, 78.14658



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.60678, 78.14652

